CANDIDATE CASE STUDY (to be shared along with dataset)

You've joined a city bike-sharing company. The CEO stops you in the corridor and says: "Riders keep complaining they can't find a bike when they need one. Meanwhile I see bikes sitting unused all over the city. Should we just buy more bikes?" Assume the company asks you to investigate this properly over the next few weeks. Analyze the data attached:

- Run quality checks and build features to explore and understand the data
- State one testable hypothesis and explain how you would test it.
- Build a model to estimate how many bikes are needed
- Design a prototype to help the operations manager use the model output through natural language questions for understanding demand

Share the code and the final results. You can do part of the problem or the complete one. You will be questioned on your assumptions, feature understanding design of experiment and final recommendation.